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Original Research

KATMed: a knowledge-augmented transformer model for contraindication-aware medication recommendation in comorbidities

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ABSTRACT

Drug-disease contraindications in comorbidities (DDCC) pose a significant challenge and priority in clinical treatment. These contraindications exhibit a prototypical long-tail distribution, characterized by low-frequency, highly diverse, and substantial individual variability. Such distinct properties impose significant limitations on electronic health record-based medication recommendation modeling, ultimately elevating safety risks in clinical practice. To address this challenge, this study proposes KATMed, a knowledge-augmented transformer model for contraindication-aware medication recommendation in comorbidities. The model employs Transformer-based encoding of patient records and leverages two self-supervised tasks to capture rich temporal and semantic dependencies. Based on this foundation, a hybrid knowledge-augmented framework is developed to integrate bidirectional medication-related clinical associations. Positive disease-procedure associations are modeled by using a dynamic semantic relevance matrix to expand the input information, thereby enhancing the model's feature learning capability on sparse yet diverse comorbidity records. Negative DDCC rules are incorporated as differentiable logical constraints in the loss function to suppress unsafe medications. Experiments on the MIMIC-III and MIMIC-IV datasets show that KATMed significantly improves performance, achieving a 5.2% increase in accuracy and a 2.04% reduction in safety violations.

1. Introduction

In modern healthcare systems, medication recommendation models serve as pivotal tools for advancing precision medicine. By integrating multi-source data encompassing patient disease profiles, medical histories, and genomic information, these models generate scientifically grounded medication recommendations to assist clinicians in optimizing treatment efficacy, minimizing adverse drug reactions, and improving healthcare resource allocation[1].

Currently, most medication recommendation models are predominantly founded on data-driven paradigms[2–7], which critically depend on large-scale electronic health records (EHRs) including structured disease codes, medication histories, and laboratory results. However, such large-scale EHRs often suffer from heterogeneity across institutions, inconsistent documentation practices[4], and intrinsic noise. A recent analysis of the MIMIC-IV dataset also revealed that 32% of medication orders exhibit discrepancies with disease codes[8], highlighting the challenges of deriving reliable patterns from raw EHRs. Moreover, purely data-driven paradigms may inadequately capture pharmacodynamic mechanisms, relying instead on superficial correlations derived from raw EHRs. For instance, models trained on observational data may falsely associate painkillers with hypertension due to coincidental co-prescription, neglecting the causal relationship between chronic pain and stress-induced blood pressure elevation[9]. Therefore, data- and knowledge co-driven paradigms are gaining attention. Recent advances in knowledge-augmented healthcare AI have demonstrated the critical role of clinical knowledge integration in medication recommendation modelling[10–12]. Various clinical associations, primarily binary relationships such as disease–drug indications and drug–drug synergies, have been integrated into medication

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recommendation models by using knowledge graph embeddings[10], attention mechanisms[13], etc. These developments are also comprehensively reviewed in recent surveys on multimodal EHR-based clinical predictive modeling and knowledge-integrated foundation models [41–44], which further emphasize the need to jointly exploit heterogeneous EHR data and structured clinical knowledge for robust and safe decision support.

Drug-disease contraindications in comorbidities (DDCC) refers to scenarios where a medication provides therapeutic benefits for one condition but exacerbates risks in a coexisting disease. For an example, β-blockers can effectively manage hypertension but may induce bronchospasm in asthmatic patients. These contraindications exhibit a prototypical long-tail distribution, characterized by low-frequency, highly diverse, and substantial individual variability. Fig. 1 gives DDCC analysis in the MIMIC-III dataset. As shown in this figure, 23.7% of patients with comorbidities (more than 2 chronic conditions) encountered at least one DDCC scenario, involving 147 high-risk drug-disease pairs across 12 therapeutic categories. For each drug-disease pair, the proportion of affected population is relatively small. However, when considering all high-risk drug-disease combinations collectively, the cumulative proportion becomes clinically significant at 23.7%. Although this still represents less than one-quarter of the total population, these cases are crucial for ensuring medication safety. Research indicates that unaddressed contraindications account for 17–26% of preventable adverse drug events in comorbid populations[14].

This long-tail distribution of contraindications poses significant challenges for EHRs-based medication recommendation modeling The purely data-driven models often fail to learn effectively from such low-frequency and highly variable data. State-of-the-art data-and-knowledge co-driven models have also failed to address DDCC. Specifically, there is neither a formalized DDCC knowledge base, nor effective knowledge fusion techniques tailored for rule-based multivariate relationships inherent to DDCC.

Based on the above observations, this study proposes an innovative deep learning model, called KATMed (Knowledge- augmented transformer model for medication recommendation), for contraindication-aware medication recommendation in comorbidities. The main contributions are as follows:

- This study develops a novel medication recommendation model by integrating EHRs-based pre-training and Transformer encoding. Two self-supervised pre-training tasks, including ICD Sequence Prediction (ISP) and Masked Sequence Reconstruction (MSR), are designed to effectively learn comprehensive representations of clinical codes and their complex interrelationships. The model employs three types of embedding representations and a Transformer-based encoder to capture both longitudinal and cross-sectional patterns of patients. Finally, medication recommendations are generated through a multilayer perceptron (MLP) with softmax activation and multi-label margin loss, enabling accurate and safe prediction of drug combinations.
- This study proposes a hybrid knowledge-augmented framework to integrate bidirectional medication-related clinical associations for improving the model performance. Positive disease-procedure associations derived from clinical guidelines and medical textbooks are modeled by using a dynamic semantic relevance matrix to expand the input information, thereby enhancing the model's feature learning capability on sparse yet diverse comorbidity records. Compared to traditional ontology-based graph embedding methods, which often suffer from semantic ambiguity and high computational cost, this matrix-based method offers a more lightweight and efficient way to encode verified clinical co-occurrence patterns. Negative DDCC rules extracted from clinical guidelines and the FDA Adverse Event Reporting System (FAERS) are incorporated as differentiable logical constraints in the loss function to suppress unsafe medications. Unlike traditional Drug-Drug Interaction (DDI) knowledge integration methods based on graph neural networks, fixed penalty terms or attention mechanisms— which are not well-suited for expressing rule-based and multivariate relationships, this constraint learning-based method effectively integrates ternary relationships of DDCC rules into the Transformer model. achieves the integration of DDCC rules based on based on ternary relationships. The integration of both types of knowledge significantly boosts model performance in terms of accuracy and safety.
- This study tested on two publicly available real-world datasets, MIMIC-III [14] and MIMIC-IV[15]. The experiments show that

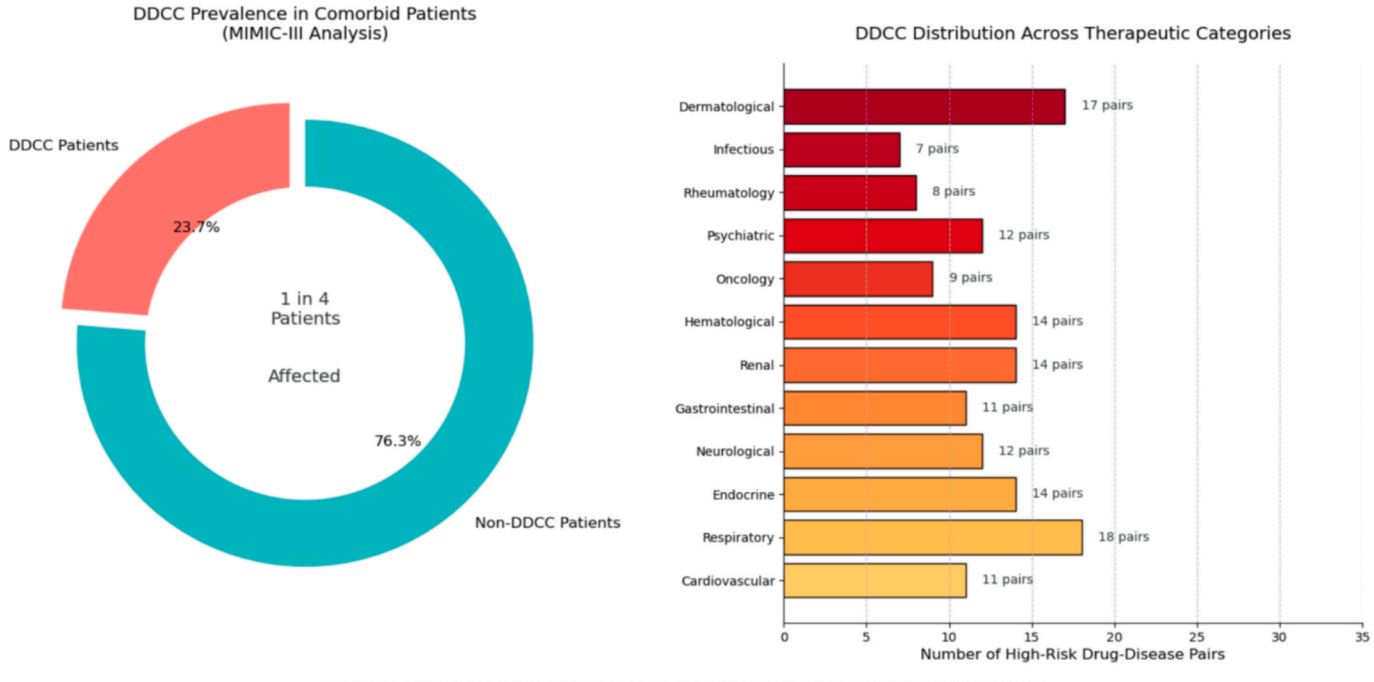


Fig. 1. Prevalence and therapeutic category distribution of DDCC in the MIMIC-III dataset.

compared with the benchmark models, this model reduces the safety risk by 2.04% and improves the accuracy by 5.2%.

2. Related works

Deep learning-based data-driven methods played a crucial role in advancing medication recommendation research. Early models typically relied on single-visit EHRs to predict medications. For example, Caballero Barajas and Akella[16] proposed a framework that used features extracted from single-visit EHRs to predict readmission risk. Their model incorporated demographic information and clinical codes from EHRs, without leveraging temporal information from prior admissions.

However, disease progression in real-world clinical settings often spans multiple visits and exhibits complex temporal patterns. To capture these longitudinal dependencies, a growing number of studies have turned to multi-visit EHR modeling. Baytas et al.[17] proposed the Time-aware LSTM (T-LSTM), which incorporates a decay mechanism into the memory cell updates of LSTM to explicitly handle irregular time intervals between visits. This design preserves temporal relevance in sequential data and improves continuity in patient trajectory modeling. Similarly, Gao et al.[18] developed StageNet, which introduces a stage transition module that encodes disease progression stages as latent representations. By jointly learning these latent stages and integrating them into the prediction module, the model effectively captures both short-term temporal dependencies and long-term disease evolution patterns[43]. These purely data-driven models demonstrated significant advantages over traditional statistical methods by learning complex sequential patterns directly from raw EHRs, and have been systematically summarized in a recent survey on multimodal EHR-based clinical predictive modelling[41].

Nevertheless, these data-driven models struggled with intricate feature interactions and noise in EHRs due to limitations, such as feature independence assumptions, overfitting, and parameter sensitivity. To address the limitations of purely data- driven models, researchers shifted toward knowledge-augmented approaches, involved with positive or negative clinical associations. Positive clinical associations, such as disease-drug indications, disease-procedure associations, and treatment compatibilities for comorbidities, have been widely used to enhance the quality of medication recommendations. They are typically embedded into deep learning models by using knowledge graph embeddings, semantic attention, or ontology-based encoding. Yang et al.[19] proposed the SafeDrug model, which focuses on generating safe and effective medication combinations. It integrated drug indication knowledge using molecular graph encoders to enhance medication safety. Wu et al.[20] developed the MEGACare framework, aiming to model complex medical relationships. It fused concepts from medical ontologies and clinical texts into a unified hypergraph representation to guide medication recommendation. Yang et al.[21] introduced MICRON, a knowledge-enhanced model built on implicit co-prescription patterns. It used historical prescription records to construct soft positive constraints for improving clinical relevance in recommendation output. However, most of these methods rely on graph-structured ontologies or static knowledge bases, which are often susceptible to outdated relations, semantic ambiguity, and high computational costs[45].

Negative clinical associations refer to contraindicated relationships between drugs and other clinical entities that may lead to adverse effects. These include DDI, DDCC, condition-specific prescription conflicts, and so on. Unlike positive associations that describe therapeutic compatibility, negative associations signal safety risks and must be strictly avoided. To ensure medication safety, negative clinical associations are often more critical than positive ones in the context of medication recommendation. In recent years, researches have increasingly focused on embedding such safety constraints into recommendation models. Most prior work concentrates on DDI, typically using graph neural networks, fixed penalty terms or attention mechanism to encode interaction risks. For example, Shang et al.[22] proposed the GAMENet model, which enhances memory-augmented neural networks with DDI graphs. This structure enables the model to suppress hazardous drug combinations during learning. Kuntal et al.[23] introduced CompNet, which embeds DDI knowledge by incorporating predefined interaction penalties into the training objective, aiming to reduce unsafe prescriptions. Yang et al.[19] introduced SafeDrug, which integrates DDI knowledge via an attention mechanism that balances medication effectiveness and safety. This design allows the model to learn optimal drug combinations while minimizing adverse interactions.

Despite these advances, existing methods are limited to binary DDI modeling and overlook the rule-based multivariate relationships of contraindications arising from comorbidities. According to recent studies[24], over 30% of hospitalized patients have at least two chronic conditions, and comorbidity-related prescription errors account for a significant portion of adverse drug events. Therefore, DDCC are more clinically relevant for safety-sensitive recommendations than DDI. However, comorbidities are highly prevalent in clinical settings. These complex conditions may involve multiple interacting diseases, leading to complex, multi-way contraindications that are difficult to represent by using conventional graphs, fixed penalty terms or attention mechanisms, which are not well-suited for expressing rule-based multivariate relationships. Therefore, current models rarely integrate DDCC knowledge comprehensively, resulting in substantial safety gaps, particularly for patients with complex comorbidities. In this context, KATMed can be viewed as a specialized instance of knowledge-integrated EHR modelling[41,44] that focuses specifically on drug–disease contraindications under comorbidity and incorporates explicit DDCC-based safety constraints into both representation learning and optimization.

To address these limitations, this study proposes a novel medication recommendation model based on a Transformer architecture. A hybrid knowledge-augmented framework are developed to integrate positive disease-procedure associations and negative DDCC rules by using the the dynamic semantic relevance matrix and constraint learning, respectively.

3. Preliminaries

Definition 1. ***(Electronic Health Records)***. *The patient's EHRs can be expressed as an ordered set of variables:* $\theta^{(n)} = \{R_1^{(n)}, R_2^{(n)}, \cdots, R_{T^{(n)}}^{(n)}\}$, *n represents the n-th patient in the sequence,* $T^{(n)}$ *represents the number of visits of the n-th pa-tient. In this study, each EHR* $R_i^{(n)}$ *for one patient's visit contains three medical codes and can be represented as* $R_i^{(n)} = \{d_i^{(n)} \cup m_i^{(n)} \cup p_i^{(n)}\}$, *which is composed of a disease code* $d_i^{(n)}$, *a medication code* $m_i^{(n)}$ *and a procedure code* $p_i^{(n)}$.

Definition 2. ***(Medication Recommendation)***. *For the specified* $R_t^{(n)}$ *of the t-th visit of the n-th patient, the disease code is* $d_t^{(n)}$, *and the medical history is* $\theta^{(n)} = \{R_1^{(n)}, R_2^{(n)}, \cdots, R_{t-1}^{(n)}\}$. *Medication recommendation is the task of predicting a multi-label output* $\widehat{y}_t \in \{0,1\}^{|M|}$, *where* $|M|$ *denotes the number of medication codes, based on the current disease code* $dt^{(n)}$, *associated symptoms, and the patient's historical visits* $\theta^{(n)}$. *The prediction output is a personalized set of medications appropriate for the patient's current condition, thus transforming the recommendation into a multi-label classification problem.*

4. Bidirectional medication-related clinical associations

4.1. Disease-procedure associations

Disease–procedure associations represent a form of positive clinical correlation reflecting standard or guideline–recommended procedures applicable to specific diseases. Fig. 2 presents an example of a disease–procedure association from the guideline section titled *Medical*

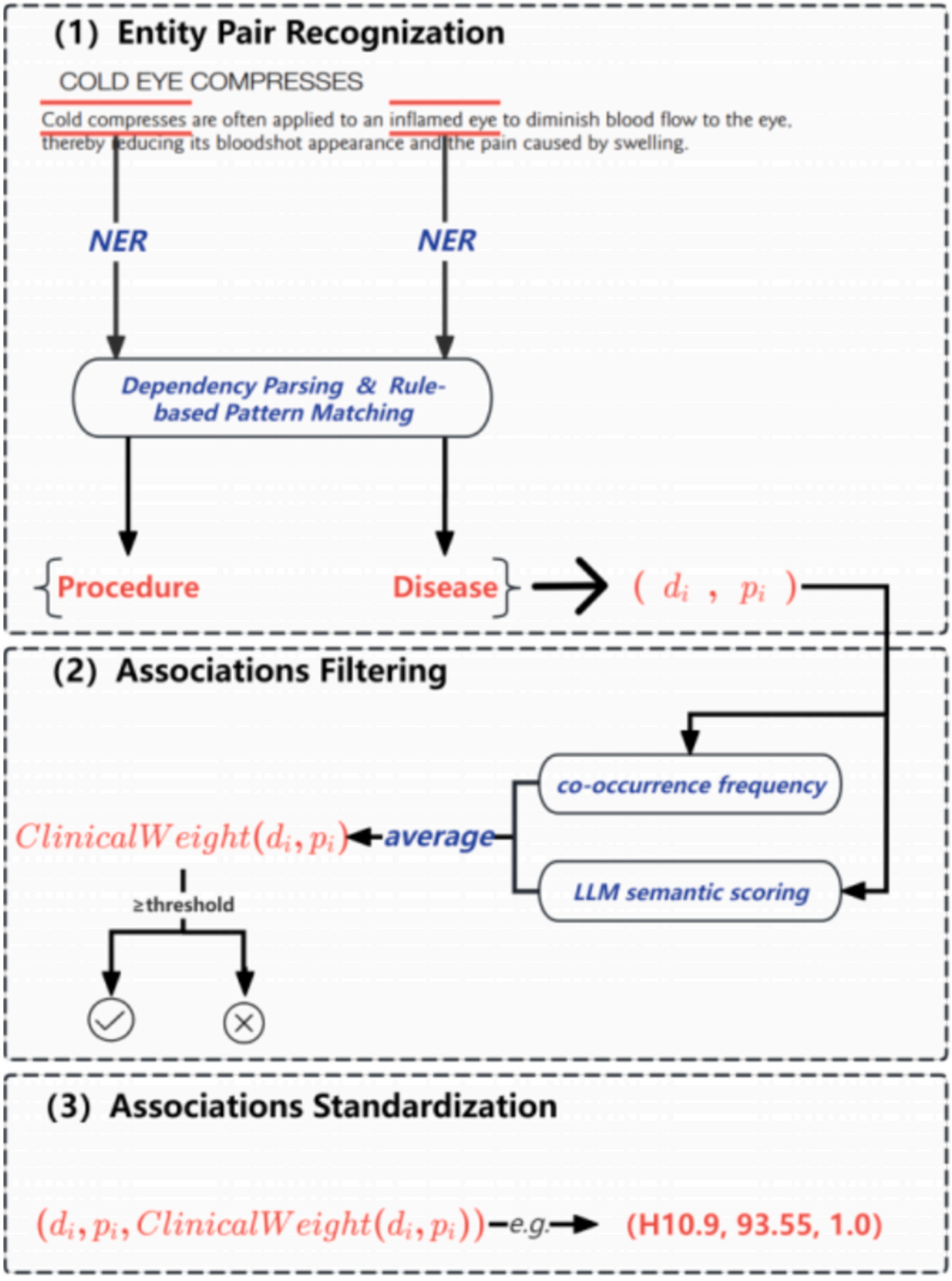


Fig. 2. Illustration of the pipeline for extracting disease–procedure associations.

Procedures–Applying Cold[25], which explicitly describes therapeutic procedures of cold compresses and their recommended use for inflammation or hemorrhage control. As shown in this example, the disease "inflamed eye" and the procedure "cold compress" form a disease–procedure association. These associations are collected from multiple authoritative sources, including the *Clinical Pathways for Common Diseases*[25] the *World Health Organization's Emergency Medical Guidelines*[26] and the *NICE Guidelines on Multimorbidity Care*[27].

This study adopts a multi–stage knowledge extraction workflow to systematically collect and structure clinically meaningful disease–procedure associations. As shown in Fig. 2, the workflow includes three steps.

Step 1: Entity discovery and candidate pairing. Guideline and pathway documents were collected between January 2023 and April 2024 from four major sources — the Chinese Clinical Pathway Database [25], the NICE guidelines on multimorbidity care[27], the WHO Emergency Medical Guidelines[26], and Harrison's Principles of Internal Medicine[28]. This study first filtered sections containing explicit therapeutic recommendations, and then applied a clinical NER model to identify Disease (e.g., 'inflamed eye') and Procedure (e.g., 'cold compress') entities. Candidate disease–procedure pairs were generated using two simple rule patterns that link disease entities to procedure phrases when the surrounding text expresses an explicit or implicit recommendation. The concrete rule templates and additional examples are provided in Appendix A. Supplementary data (Table A1). For example, from the sentence 'Cold compresses are often applied to an inflamed eye', the candidate pair (di, pj) = ('eye inflammation', 'cold compress') is produced. Summary statistics of document collection, entity discovery, and candidate pairing are reported in Table 4.

Step 2: Association scoring and filtering. To quantify association strength, each pair is assigned a clinical weight:

$$\mathrm{ClinicalWeight}\left(\mathrm{d_i},\mathrm{p_j}\right)=\frac{1}{2}\left[\mathrm{Freq}\left(\mathrm{d_i},\mathrm{p_j}\right)+\mathrm{LLM}\left(\mathrm{d_i},\mathrm{p_j}\right)\right]\in[0.1,1.0] \tag{1}$$

The clinical weight in Eq. (1) is a combination of two normalized components: Freq(d_i,p_j) and LLM(d_i,p_j). First, Freq(d_i,p_j) is obtained by taking the logarithm of the raw co-occurrence count for the pair (d_i,p_j) over all guideline and pathway documents, followed by min–max normalization across all candidate pairs to obtain a frequency score in [0, 1]. Second, LLM(d_i,p_j) is a context-aware confidence score predicted by a large language model. For each candidate pair, the model is asked to rate the strength of the association on a five-level ordinal scale {*very strong, strong, moderate, weak, unlikely*}, which is then mapped to numeric values in [0, 1] and linearly rescaled.

The final clinical weight is computed as a weighted combination of these two components, with the weight for each component optimized on a validation set. The mixing coefficient is determined by evaluating the model's performance on a validation set, ensuring the optimal balance between the co-occurrence frequency and LLM-based confidence score. The decision threshold τ for ClinicalWeight is tuned over the range [0.1, 1.0] using a held-out validation set to balance coverage and noise removal. The final τ is then fixed for all subsequent experiments. Candidate pairs whose weight falls below $\tau\backslash tau\tau$ are discarded from further processing.

To further reduce noise, this study applies cross-document consistency checks and near-duplicate removal. Cross-document consistency ensures that each disease–procedure pair is validated across multiple source documents. Pairs found in only one document are discarded. Near-duplicate removal is based on min–hash Jaccard similarity (threshold > 0.9), where pairs with high similarity (greater than 0.9) are considered duplicates and excluded.

Step 3: Standardization and knowledge base materialization. All retained disease and procedure entities are normalized to standardized terminologies — **ICD-10** for diseases (e.g., "eye inflammation" → *H*10.9) and **ICD-9-CM** or **SNOMED CT** for procedures (e.g., "cold compress" → 93.55). Normalization relies on a deterministic dictionary built from official code lists and combines text normalization, abbreviation expansion, staged lookup (exact, lemmatized, and fuzzy matching), and ontology-guided disambiguation. Ambiguous surface forms are kept only when the guideline context uniquely supports a single code; otherwise, they are discarded to avoid label noise. Each validated association is then stored as a canonical triple as in Eq. (2) together with provenance metadata (source document, section title, and paragraph indices).

$$DPA_{i,j}=\left(d_i^{\mathrm{ICD10}},p_j^{PROC},\mathrm{ClinicalWeight}\left(d_i,p_j\right)\right)=(\mathrm{H10.9},\ 93.55,1.0) \tag{2}$$

The resulting disease–procedure association knowledge base is serialized in tabular and JSON formats and is later embedded via the dynamic semantic relevance matrix to provide the Association Embedding (E_{ass}) in the input layer of KATMed. For further details on the disease–procedure association extraction pipeline, including full lists of the crawling keywords, parsing rules, additional data related to candidate pairing, validation results and intermediate statistics, please refer to Appendix A. Supplementary data.

4.2. Drug-disease contraindications in comorbidities

Compared to positive associations, drug–disease contraindications in comorbidities (DDCC) are negative clinical associations essential for enforcing safety constraints in medication recommendation models.

Sources and scope. DDCC knowledge is collected from authoritative national and international sources, including the *2023 Guidelines for the Prevention and Treatment of Hypertension in China*[29], the *Handbook of Emergency Cardiovascular Pharmacotherapy*[30], and the *Chinese Pharmacopoeia*[31]. These documents contain structured descriptions of adverse effects and contraindication conditions suitable for reliable extraction. Table 2 summarizes examples of medications with formally documented contraindications.

Extraction pipeline and materialization. Guidelines are digitized using PaddleOCR v2.6 and Tabula for table parsing where necessary. This study identify **Drug**, **Contraindicated Conditions**, and **Adverse Effects** fields; diseases are recognized by the same NER model and mapped to **ICD–10**, while drugs are normalized to **ATC** Level–3 codes. For complex disease mentions (e.g., aortic valve stenosis, metabolic syndrome), SNOMED CT[32] expands terminology to ensure robust normalization. Following entity reorganization, multi–entity rules are compiled using dependency parsing and predefined templates:

- IF {<Disease1>, <Disease2>} THEN NOT < Drug > [Severity: absolute/relative]
- <Drug > is contraindicated in < Disease > patients [Severity: absolute/relative]
- <Drug > may worsen < Disease > and should be avoided [Severity: absolute/relative] Each instance is represented as:

Table 1
Statement of Significance.

Problem or Issue	Patients with multiple chronic conditions face elevated risks of drug–disease contraindi- cations (DDCC), which follow a long-tail distribution and are poorly captured by existing data-driven medication recommendation models, leading to preventable adverse drug events.
What is Already Known	Recent knowledge-augmented models (e.g., SafeDrug, MEGACare) improve prediction accuracy by integrating drug-drug interactions or ontologies, but most fail to formally encode comorbidity-specific DDCC rules and lack explicit safety supervision.
What this Paper Adds	This study proposes **KATMed**, a knowledge-augmented Transformer framework that integrates positive disease–procedure associations and negative DDCC rules, combined with dual self-supervised pretraining and differentiable safety constraints, to achieve state-of-the-art accuracy and reduced contraindication violations.
Who Would Benefit	Clinicians seeking safer AI-assisted prescribing tools, biomedical informatics researchers developing interpretable medication recommendation models, and health systems aiming to minimize preventable adverse drug events in comorbid populations.

Table 2
Examples of drug–disease contraindications and adverse effects (adapted from[29]).

Drug	Therapeutic Class (ATC)	Adverse Effects	Contraindicated Conditions (ICD–10)
Diltiazem	C08DB01 (Calcium channel blockers)	Bradycardia, AV block, hepatotoxicity, constipation	Severe hypotension or cardiogenic shock, 2nd–3rd degree AV block (I44. 1), sinus node dysfunction, severe heart failure (I50. 1)
Nitroglycerin	C01DA02 (Vasodilators)	Headache, vomiting	Nitrate allergy, severe anemia, intracranial hypertension (G93.2), angle–closure glaucoma, hypotension (I95. 0)
Metformin	A10BA02 (Biguanides)	GI discomfort, rare lactic acidosis	Hypoxemia (R09. 02), hepatic insufficiency (K72. 0), heart failure (I50. 1)
β–blockers	C07AB02 (β–blocking agents)	Bradycardia, fatigue, cold extremities, depression	Asthma (J45. 9), bradycardia (R00. 1), AV block (I44. 1)
Labetalol	C07AG01 (α and β blockers)	Nausea, headache, bronchospasm, orthostatic hypotension	Bronchial asthma (J45. 9), cardiogenic shock (R57. 0), AV block (I44. 1), severe heart failure (I50. 1)
Esmolol	C07AB09 (β–blockers)	Hypotension, nausea	Severe sinus bradycardia (R00. 1), AV block (I44. 1), decompensated heart failure (I50. 1)
Nicardipine	C08CA04 (Dihydropyridine CCB)	Tachycardia, headache, edema	Severe aortic stenosis (I35. 0), cardiogenic shock (R57. 0), acute MI (I21. 9)
Urapidil	C02CA06 (α–1 adrenergic blockers)	Hypotension, dizziness, nausea, tiredness	Aortic isthmus stenosis, AV shunt (except during dialysis) (Q25. 1)

$$\mathrm{IF}\{d_i, d_j, \ldots\} \Rightarrow \neg\, m_k \quad [\mathrm{Severity} : s], \tag{3}$$

where *di* are ICD–10 codes, m_k is an ATC code, and $s \in$ {absolute, relative} is a severity label automatically assessed by a large language model given the contraindication context and evidence. This label controls the penalty strength during model training without changing experimental protocols. Examples derived from Table 2 include:

$$\mathrm{IF}\{\mathrm{R00.1}, \mathrm{I44.1}\} \Rightarrow \neg \mathrm{C08DB01} \quad [\mathrm{Severity} : \mathrm{absolute}] \tag{4}$$

$$\mathrm{IF}\{\mathrm{I50.1}\} \Rightarrow \neg \mathrm{C08DB01} \quad [\mathrm{Severity} : \mathrm{relative}] \tag{5}$$

$$\mathrm{IF}\{\mathrm{G93.2}, \mathrm{I95.0}\} \Rightarrow \neg \mathrm{C01DA02} \quad [\mathrm{Severity} : \mathrm{absolute}] \tag{6}$$

Quality control and reproducibility. Ambiguous DDCC entries are resolved through a fully automated pipeline without manual clinical review. First, all ICD–10/ATC/SNOMED CT codes are validated against version–pinned code lists; invalid or deprecated codes are rejected or mapped forward when an official forward map exists. Second, rules are canonicalized to the schema in Eq. (3) with deterministic key ordering of disease sets (lexicographic by code) to enable exact duplicate removal. Third, contradictions are detected by a constraint auditor: (i) a DDCC rule that permits a drug under a stricter disease superset is flagged; (ii) within–drug severity conflicts ("absolute" vs. "relative") across sources are reconciled by a precedence policy that prefers the most recent guideline and then the stricter label; ties default to "absolute" for safety. Fourth, severity labels produced by the large language model are cross–checked by a two–pass self–consistency procedure (temperature ∈ {0.2, 0.8}); labels with disagreement are downgraded to "relative" unless corroborated by an explicit "contraindicated" cue in the source sentence. Finally, per–rule provenance (document ID, section, page/line, extraction template) is retained for auditability.

LLM configuration and robustness checks. Both the LLM-based confidence scores $\mathrm{LLM}(d_i, p_j)$ used in Eq. (1) and the severity labels s in Eqs. (4)–(6) are generated by GPT-4o-mini. For each disease–procedure pair, the model is prompted with the local guideline or textbook context and asked to assign a five-level ordinal rating of association strength; for each candidate DDCC rule, it is asked to assign a categorical label indicating whether the contraindication is absolute or relative. The exact prompt templates and illustrative examples are provided in Appendix A. Supplementary data (Table A2). Decoding is performed with a low temperature and a short maximum generation length to encourage deterministic, classification-style outputs. For each disease–procedure pair or drug–disease rule, two independent LLM responses are obtained; only pairs/rules with consistent labels are retained, and a clinician manually reviews a random subset to verify plausibility and internal consistency. All subsequent experiments use the resulting scores and labels as fixed inputs.

The resulting DDCC repository materializes 312 validated rules (covering 428 ICD–10 disease codes and 153 ATC Level–3 classes) in a machine–readable JSON/CSV format aligned with Eq. (3). Integration into the model is performed as differentiable safety constraints in the loss (Eq. (21)–(23)). All scripts fix random seed to 2024; library versions, code lists, and ontology snapshots are pinned to ensure bitwise–reproducible builds.

5. Methods

This study proposes KATMed, a knowledge-augmented medication recommendation model that integrates bidirectional medication-related clinical associations into a Transformer backbone.The framework of our KATMed model is shown in Fig. 3. Methodologically, our main contributions lie in: (i) constructing and embedding disease–procedure associations and drug–disease contraindications (DDCC) into a unified representation space; (ii) designing a constraint-aware learning objective that explicitly penalizes DDCC-violating recommendations; and (iii) introducing dual self-supervised pretraining tasks to better exploit longitudinal EHR signals. Other components of the pipeline—including standard EHR preprocessing, visit-level sequence construction, tokenization, and optimization—follow widely used practices in prior work and are therefore only summarized briefly in the main text; detailed, step-by-step descriptions are provided in Appendix A (Online Resource 1).

5.1. Embedding module incorporating disease-procedure associations

The embedding module is designed to convert raw EHRs—defined in Definition 1 as ordered sets of medical codes per visit—into unified, low-dimensional representations that capture both temporal and semantic dependencies among clinical events. Each patient's longitudinal EHR sequence is modeled as a series of visits, where each visit contains disease, procedure, and medication codes standardized under ICD-9-CM or ATC mappings.

5.1.1. Scope and coding standards

Following Definition 1, each visit is represented by structured disease, procedure, and medication codes extracted from MIMIC-III and MIMIC-IV. Diagnoses are mapped to ICD-9/ICD-10, procedures to ICD-9-CM/SNOMED CT, and medications to ATC Level-3/Level-4 codes using official code lists and standard cross-walks. Rare or deprecated codes are

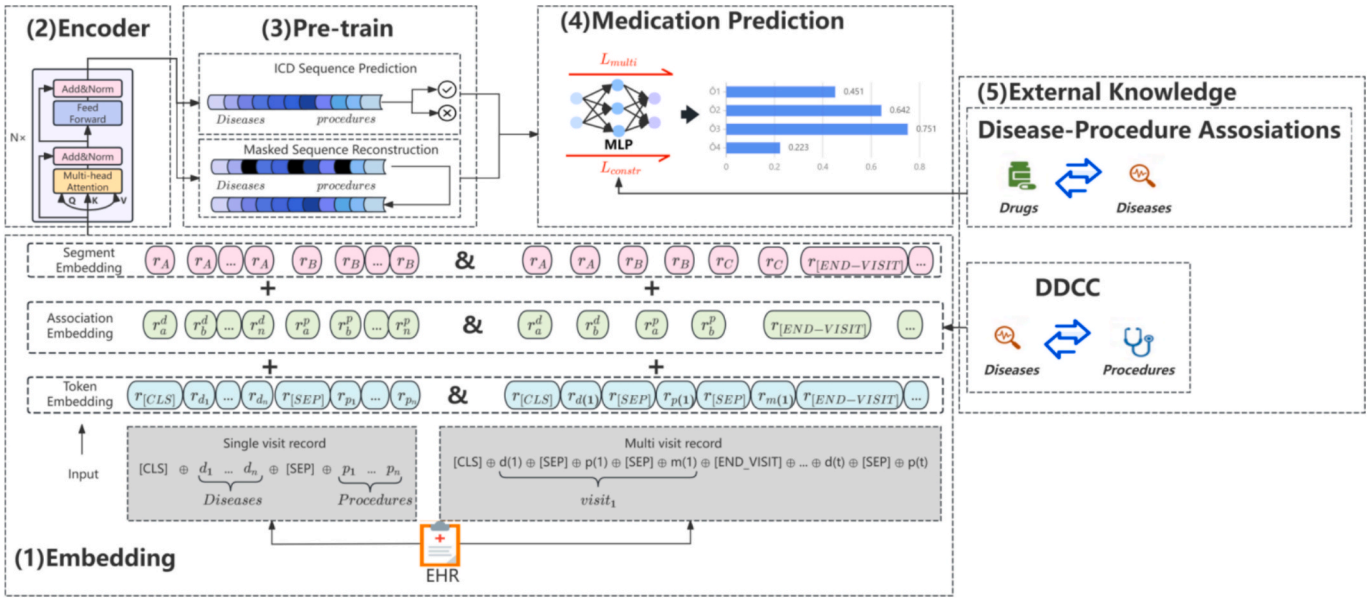


Fig. 3. The framework of the proposed KATMed.

merged or forward-mapped to their contemporary equivalents, and visits with no valid clinical codes are discarded. These preprocessing and coding choices follow common practice in EHR-based medication recommendation studies; the full mapping tables, frequency thresholds, and handling rules are documented in Appendix A.7 (Online Resource 1) for transparency and reproducibility.

5.1.2. Single-visit scenario: Preprocessing and structuring

In the single-visit setting (T = 1), each admission is treated as one visit. All disease and procedure codes assigned during the admission are aggregated into de-duplicated multi-hot sets, and the medication set for that visit is used only as the prediction target. No longitudinal context or temporal gap information is introduced in this case, so the model focuses on learning the mapping from current diagnoses/procedures to medications. The concrete preprocessing steps (visit unit definition, code aggregation and cleaning rules, and input sequence templates for single-visit records) are summarized in Appendix A.7, Table A7 (Online Resource 1).

5.1.3. Multi-visit scenario: Preprocessing and structuring

In the multi-visit setting (T ≥ 2), all admissions of a patient are first sorted chronologically to form a visit sequence ($v_1,\ldots,v_T$). For each visit, diagnosis and procedure codes are aggregated into sets as in the single-visit case, while medication codes from previous visits ($m_1,\ldots,m_{t-1}$) are treated as contextual input features. The medication set at the current visit m_t is never included in the input and is used solely as the label y_t, thereby preventing label leakage. Irregular time intervals Δt between visits are encoded as learnable time-gap embeddings, and sequences longer than a maximum length T_{max} are truncated to the most recent visits.

5.1.4. Input sequence construction

The final input sequence is constructed in chronological order by interleaving diagnosis, procedure, and (historical) medication tokens, separated by special markers such as [CLS],[SEP],[END_VISIT] and [PAD] (see Eqs. (7)–(8)). This BERT-style sequence construction follows standard practice in Transformer-based EHR modeling; a step-by-step description, including sequence templates, padding/truncation schemes, and token type encodings, is provided in Appendix A.7–A8, Tables A7–A8 (Online Resource 1).

$$\theta = \{R_1, R_2, \cdots, R_T\} = \{\{d_1 \cup m_1 \cup p_1\}, \cdots, \{d_T \cup m_T \cup p_T\}\} \tag{7}$$

$$\begin{aligned}\text{input} = {} & [\text{CLS}] \oplus d(1) \oplus [\text{SEP}] \oplus p(1) \oplus [\text{SEP}] \oplus m(1) \oplus [\text{END_VISIT}] \\ & \oplus \cdots \\ & \oplus d() \oplus [\text{SEP}] \oplus p(t-1) \oplus [\text{SEP}] \oplus m(t-1) \oplus [\text{END_VISIT}] \oplus d(t) \oplus [\text{SEP}] \\ & \oplus p(t)\end{aligned} \tag{8}$$

5.1.5. Embedding components

To enhance representational ability, KATMed introduces three embedding components—Token Embedding, Segment Embedding, and Association Embedding— each capturing a complementary aspect of clinical semantics.

5.1.6. Token embedding

(*E*tok) converts each medical code token *xi* into a dense vector via a trainable matrix *W*tok ∈ $\mathbb{R}^{|V|\times d}$, initialized with Xavier[33] scheme:

$$\mathrm{E}_{\mathrm{tok}}(\mathrm{x}_\mathrm{i}) = \mathrm{W}_{\mathrm{tok}}[\mathrm{x}_\mathrm{i}] \tag{9}$$

5.1.7. Segment embedding

(*E*seg) indicates the semantic type of each token. Each token has a type indicator *si* ∈ {0, 1, 2}, where 0, 1, and 2 correspond to disease, procedure, and medication codes, respectively. A segment embedding matrix $W_{\mathrm{seg}} \in \mathbb{R}^{3\times d}$ is initialized from a uniform distribution U (—0.01, 0.01) and trained jointly:

$$E_{\mathrm{seg}}(x_i) = W_{\mathrm{seg}}[s_i] \tag{10}$$

5.1.8. Association embedding

(E_{ass}) integrates prior disease–procedure associations through a matrix $A \in \mathbb{R}^{|d|\times|p|}$:

$$A_{i,j} = \alpha\cdot\mathrm{ClinicalWeight}\left(d_i, p_j\right) + (1-\alpha)\cdot\mathrm{SemanticSim}\left(d_i, p_j\right), \tag{11}$$

where α ∈ [0, 1] balances clinical co-occurrence weights and semantic similarity. SemanticSim(*di, pj*) is computed as the cosine similarity between textual embeddings of diseases and procedures. The matrix is projected into token-level vectors via trainable matrices *Wd* and *Wp*:

$$E_{\mathrm{ass}}(d_i) = \sum_{j=1}^{|p|} A_{i,j}\cdot W_p\left[p_j\right], \quad E_{\mathrm{ass}}\left(p_j\right) = \sum_{i=1}^{|p|} A_{i,j}\cdot W_d[d_i] \tag{12}$$

Finally, all embeddings are combined element-wise and passed

through the Transformer encoder to obtain the patient-level contextual representation:

$$r = \text{Transformer}\left(E_{tok}(x) + E_{seg}(x) + E_{ass}(x)\right) \tag{13}$$

This unified embedding design ensures that both temporal dependencies and knowledge-driven semantic relationships are preserved across single-visit and multi-visit scenarios, supporting reproducible and clinically consistent modeling of longitudinal EHRs.

5.2. Pre-training module with dual-task binding

Pre-training module is designed to enhance the model's in-depth understanding of clinical codes. KATMed employs two self-supervised pre-training tasks: ICD Sequence Prediction (ISP) and Masked Sequence Reconstruction (MSR). By exploring contextual dependencies and latent semantic associations among clinical codes, these two tasks generate robust feature representations for downstream medication recommendation.

ISP Task: The goal of this pre-training task is to determine whether the disease and procedure code sequences D and P are derived from the same patient. It helps the model to learn the contextual dependencies between different medical codes and enhances the model's ability to recognize patient-level semantic consistency. To achieve this, negative examples are generated by replacing one or more codes in D or P with codes from another patient, while positive examples are constructed from authentic records of the same patient. The model then learns to discriminate between positive and negative pairs via a binary classification task.

The predicted probability vector is obtained as:

$$\widehat{y} = \text{softmax}(W_1 r + b_1) \tag{14}$$

where r is the patient representation from the encoder, $W1$ is the weight matrix and $b1$ the bias of the classifier. The prediction $\widehat{y} \in$ R2 contains the probabilities of the input being a positive or negative sample.

The loss function is binary cross-entropy:

$$L_{\text{ISP}} = -y\log\left(\widehat{y}_{\text{pos}}\right) - (1-y)\log\left(\widehat{y}_{\text{neg}}\right) \tag{15}$$

where $y \in \{0, 1\}$ is the ground-truth label, and ŷpos, ŷneg are the predicted probabilities of being positive or negative, respectively.

MSR Task: The goal of this pre-training task is to reconstruct masked codes based on surrounding context. It promotes the model's ability to capture semantic structure and code dependencies. Specifically, a subset of tokens in the input sequence is randomly masked, and the model is trained to recover these tokens.

The predicted probability vector is obtained as:

$$\widehat{s} = \text{softmax}\left(W_2 r_{mask} + b^2\right) \tag{16}$$

where r_{mask} is the hidden state at the masked position, W_2 and b_2 are trainable parameters.

The MSR loss is defined as the cross-entropy between the predicted distribution $\widehat{s}$ and the ground-truth distribution **s**, computed over the set of masked positions $i \in M$.

$$L_{\text{MSP}} = -\sum_{i \in M}\sum_{c=1}^{C} s_{i,c}\log\left(\widehat{s}_{i,c}\right) \tag{17}$$

where C denotes the total number of medical code classes, and $s_{i,c}$ is a one-hot label indicating whether the c-th class is the correct token at position i. This objective encourages the model to accurately reconstruct the masked disease, procedure or medication codes from the surrounding context.

5.3. Medication prediction module constrained by drug-disease contraindications in comorbidities

The medication prediction module aims to recommend a personalized drug combination based on the patient-level representation **r**, which is obtained from the transformer encoder pretrained via self-supervised tasks. Specifically, KATMed employs a MLP to predict the probability of each medication being included in the final prescription:

$$\widetilde{\mathbf{O}} = \text{softmax}(W_3\mathbf{r} + \mathbf{b}_3) \tag{18}$$

where $\widetilde{\mathbf{O}} \in [0,1]^{|V_m|}$ denotes the predicted medication probability vector, where $|Vm|$ is the total number of candidate medications.

r is the contextualized patient representation derived from the pretrained embedding and encoding modules. $W_3 \in \mathbb{R}^{|V_m| \times d}$ and $\mathbf{b}_3 \in \mathbb{R}^{|V_m|}$ are learnable parameters of the prediction layer.

To train this module, a multi-label margin loss is adopted to ensure that the predicted probabilities for ground-truth medications exceed those of irrelevant ones:

$$L_{multi} = -\sum_{i,j:\mathrm{m_i}=1,m_j=0} \frac{max\left(0, 1 - \left(\widetilde{\mathbf{O}}_i - \widetilde{\mathbf{O}}_j\right)\right.}{|V_{\mathrm{m}}|} \tag{19}$$

where $\mathbf{m} \in \{0,1\}^{|V_m|}$ is a binary indicator vector denoting the true medication labels for the patient; specifically, $m_i = 1$ means the i-th drug is prescribed, and $m_j = 0$ indicates the j-th drug is not. $\widetilde{\mathbf{O}} \in [0,1]^{|V_m|}$ is the predicted probability distribution over the entire drug vocabulary, where $\widetilde{\mathbf{O}}_i$ is the predicted score of the i-th drug. The margin term max $\left(0, 1 - \left(\widetilde{\mathbf{O}}_i - \widetilde{\mathbf{O}}_j\right)\right)$ penalizes cases where the predicted score for a relevant drug is not sufficiently greater than that of an irrelevant one. $|V_m|$ denotes the total number of candidate medications and is used to normalize the overall loss across all positive–negative drug pairs.

5.3.1. Contraindication constraints

DDCC rules are incorporated into this medication prediction module to constraint model training. This study adopts constraint learning to achieve this knowledge integration. By using the product norm, these DDCC rules are transform into differentiable functions, and use the minimization of the differentiable functions to constrain the objectives. Specifically, if a patient simultaneously presents with a comorbidity pair (V_{d1}, V_{d2}), then a known contraindication V_m should be discouraged from being prescribed. The logical constraint is expressed as:

$$\bigwedge_{(V_{d1},V_{d2},V_m)\in \text{DDCC}} \Delta(V_{d1}, V_{d2}) \rightarrow o(V_m) \tag{20}$$

To enforce this rule in a differentiable way, this study define the following loss:

$$L_{\text{constr}} = \lambda \cdot \sum_{(V_{d1},V_{d2},V_m)\in \text{DDCC}} \Delta(V_{d1}, V_{d2}) \cdot \log(1 - o(V_m)) \tag{21}$$

where $\Delta(V_{d1}, V_{d2})$ is an indicator function returning 1 if both V_{d1} and V_{d2} are present in the patient record, and 0 otherwise. $o(V_m)$ is the predicted probability of recommending medication V_m. The constraint encourages the model to suppress contraindicated drugs under high-risk comorbidities. $\lambda \geq 0$ is a hyperparameter that controls the strength of this constraint.

The final training loss jointly considers prediction accuracy and safety:

$$L = (1\text{-}\beta) \cdot L_{\text{multi}} + \beta \cdot L_{\text{constr}} \tag{22}$$

where $\beta \in [0, 1]$ balances the importance of the multi-label margin loss L multi and the safety constraint L constr.

During inference, medications with predicted probability exceeding

a fixed threshold $\mu = 0.5$ are selected as the final recommendation set:

$$\mathscr{A}_{\text{final}} = \{i | \widetilde{\mathbf{O}}_i > 0.5, 1 \le i \le |V_m|\} \quad (23)$$

6. Experiments

6.1. Datasets and knowledge base

This study conducted experiments on two benchmark datasets, MIMIC-III and its updated version MIMIC-IV. MIMIC-III includes EHRs from approximately 61,000 ICU patients between 2001 and 2012, while MIMIC-IV extends this with 46,000 ICU patient records from 2013 to 2019.

Following prior studies[19,22,34], This study mapped drug codes from National Drug Code (NDC) to **ATC** level-3 and standardized disease and procedure codes using **ICD-9**. The datasets were split into training/validation/test sets with a ratio of 4:1:1. Detailed statistics are shown in Table 3.

To further assess cross-institution generalization, this study additionally construct an external validation cohort from the eICU Collaborative Research Database, a multi-center critical care dataset covering over 200 hospitals in the United States. For consistency with the MIMIC cohorts, this study restrict eICU to adult ICU stays (age $\geq$ 18 years) with at least one chronic comorbidity, available diagnosis codes and medication records, and sufficient temporal information to form visit-level sequences. Diagnosis and procedure codes are mapped to the same ICD-based scheme used for MIMIC, and medication records are mapped to ATC Level–3 codes whenever possible. Basic statistics of the resulting eICU cohort, including the number of patients and visits, code vocabularies, and per-visit code counts, are summarized in Table 3.

This paper constructs two structured knowledge bases: positive disease-procedure associations and Negative Drug-Disease Contraindications.

6.1.1. Positive Disease-Procedure associations

A total of 8,281 clinical guideline and pathway documents were collected between January 2023 and April 2024 from *the Chinese Clinical Pathway Database*[25], *NICE Guidelines*[27], *World Health Organization's Emergency Medical Guidelines*[26], and *Harrison's Principles of Internal Medicine* (20th ed.).

Web crawling was conducted using BeautifulSoup4.12.2, and PDF parsing via pdfplumber0.10.3, restricted to documents containing the keywords "procedure", "treatment", "therapy", "recommendation", "clinical pathway", and "indication".

After parsing, a total of 7,853 HTML pages and 428 PDF files were analyzed, producing 5,412 unique disease entities and 3,872 unique procedure entities during Step 1. The candidate pairing stage (Step 2) generated 42,658 raw disease–procedure pairs, which were refined into 10,234 validated associations after scoring, standardization, and duplicate removal (Step 3).

Table 4 summarizes the construction statistics of the pipeline, including data collection, entity discovery, and pairing refinement.

Table 3
Statistics of the processed EHR datasets.

Statistic	MIMIC-III	MIMIC-IV	eICU Cohort
# of visits / # of patients	14, 842/6, 211	16, 203/6, 461	200,859/ 139,367
# of unique disease / procedures / medications	2,030/621/ 115	3,520/ 1,172/125	3,803/2,209/ 131
Avg / Max # of visits per patient	4.94/28	8.36/51	9.21/47
Avg / Max # of disease per visit	13.81/40	12.45/40	12.94/34
Avg / Max # of procedures per visit	4.31/26	2.21/26	3.22/21
Avg / Max # of medications per visit	19.22/51	12.51/53	17.92/43

Table 4
Construction and composition of the disease–procedure association knowledge base.

Item	Description	Count / Value
Data collection window	January 2023 – April 2024	—
Source repositories	Chinese Clinical Pathway Database[25], NICE Guidelines[27], World Health Organization's Emergency Medical Guidelines[26], Harrison's Principles of Internal Medicine (20th ed.)	4 main sources
# of source documents crawled	Total documents (HTML + PDF)	8,281 (7,853 HTML + 428 PDF)
Named entity categories	Disease entities / Procedure entities	5,412 / 3,872
Candidate pairs generated (Step 2)	Raw disease–procedure pairs before filtering	42,658
Validated associations (Step 3)	Final filtered disease–procedure pairs	10,234
Avg. ClinicalWeight score	Weighted co-occurrence + semantic score	0.72 ± 0.11
Mapping standards	ICD-10 (diseases), ICD-9-CM / SNOMED CT (procedures)	—
Quality control	Cross-source concordance $\geq$ 2 sources, duplicate Jaccard > 0.9 removed	Pass rate 95.5%

The final disease–procedure association knowledge base contains 10,234 validated and ontology-aligned (disease, procedure) pairs, covering 1,286 distinct diseases and 1,764 standardized medical procedures after normalization to ICD-10 and SNOMED CT.

6.1.2. Negative Drug-Disease contraindications

A total of 38 authoritative guideline sources were used to extract drug–disease contraindication (DDCC) rules, including *the Chinese Pharmacopoeia(2020)*[28], *2023 Hypertension Guidelines*[29], *Emergency Cardiovascular Pharmacotherapy Handbook*[31], and *SNOMED CT*[32].

Each contraindication rule was encoded as a logical constraint linking coexisting disease conditions with medications that should not be co-administered.All documents were processed between January 2023 and April 2024 using PaddleOCR v2.6 and Tabula, followed by rule extraction, normalization, and verification under standardized ontologies (ICD-10, ATC, and SNOMED CT).

During construction, the system analyzed 38 guideline documents, identified 428 unique disease entities and 153 ATC Level-3 drug classes, and produced 312 validated DDCC rules after filtering and normalization.The process statistics are summarized in Table 5, including data collection, entity recognition, and rule-validation metrics.

The finalized DDCC knowledge base contains 312 standardized and ontology-aligned contraindication rules, covering 428 ICD-10 diseases and 153 drug classes, each rule representing a verified disease–drug incompatibility that can be directly applied during medication recommendation and model evaluation.

6.2. Evaluation protocol

This study adopts widely accepted evaluation metrics[19,22,34,35] to assess the overall performance of all compared methods. The metrics include the Jaccard coefficient (Jaccard), the area under the precision-recall curve (PRAUC), the F1 score (F1), and the average number of recommended medications (#MED).

Higher values of Jaccard, PRAUC, and F1 indicate better prediction accuracy. This study also emphasize that an ideal medication recommendation should resemble a doctor's prescribing behavior: drugs should be accurately chosen, avoiding both overpre- scription and underprescription. To reflect this balance, this study include the #MED metric, which measures the average number of medications suggested by a model.

Table 5
Construction and composition of the DDCC knowledge base.

Item	Description	Count / Value
Data collection window	January 2023 – April 2024	—
Source repositories	the Chinese Pharmacopoeia(2020)[28], 2023 Hypertension Guidelines[29],, Emergency Cardiovascular Pharmacotherapy Handbook[31], SNOMED CT[32]	4 main sources
# of guideline documents parsed	Total number of national and institutional guideline documents analyzed	38
Document formats	PDF-based clinical guidelines and pharmacotherapy manuals (digitized via PaddleOCR v2.6 + Tabula)	—
Named entity categories	Disease entities (ICD-10), Drug entities (ATC Level-3)	428 diseases / 153 drug classes
Extracted DDCC rules	Validated logical contraindication rules (after normalization and filtering)	312
Avg. # of contraindicated drugs per disease pair	Average number of restricted drugs per comorbidity condition pair	2.7
Severity label distribution	Absolute vs. Relative contraindications (auto-assigned by LLM)	58% / 42%
Mapping standards	ICD-10 (diseases), ICD-9-CM / SNOMED CT (procedures)	—
Quality control	Code validation (ICD/ATC/SNOMED), cross-source reconciliation, severity self-consistency, duplicate removal (Jaccard > 0.9)	Pass rate 94.2%

All metrics are calculated as follows:

$$\text{Jaccard} = \frac{\{i : m_i = 1\} \cap \{j : m_j = 1\}}{\{i : m_i = 1\} \cup \{j : m_j = 1\}} \tag{24}$$

$$R = \frac{\{i : m_i = 1\} \cap \{j : m_j = 1\}}{\{i : m_i = 1\}}, \quad P = \frac{\{i : m_i = 1\} \cap \{j : m_j = 1\}}{\{j : m_j = 1\}}, \quad F_1 = \frac{2R \times P}{R + P} \tag{25}$$

$$\text{PRAUC} = \sum_{k=1}^{|V_m|} P_k \cdot (R_k - R_{k-1}) \tag{26}$$

$$\textbf{MED} = \sum_{i=1}^{|V_m|} \widetilde{O}_i \tag{27}$$

where $m \in \{0,1\}^{|V_m|}$ is the ground-truth multi-hot medication vetor for a patient, $\widetilde{m} \in \{0,1\}^{|V_m|}$ is the predicted medication. combination after thresholding, and $\widetilde{\mathbf{O}} \in [0,1]^{|V_m|}$ denotes the model's predicted probability vector. P_k and R_k represent the precision and recall at cutoff rank k, respectively. Note that #MED reflects the model's average prescribing tendency, and may take non-integer values. All metrics are averaged over the test set.

In addition, this study designed a novel metric, namely the Drug-Disease Contraindications in Comorbidities Violation Rate (DDCC-VR). It is the proportion of recommended drugs that violate the DDCC rules and used to evaluate the recommendation safety in comorbidities.

$$V_{ij} = \sum_{m_k \in M_{ij}} \mathbb{I}(\text{recommended drugs include } m_k) \tag{28}$$

$$\text{DDCC} - \text{VR} = -\frac{1}{|C|} \sum_{(d_i,d_j) \in C} \frac{V_{ij}}{N_{ij}} \tag{29}$$

where C is the set of all combined disease pairs (unordered combinations). $V_{i\ j}$ represents the violation frequency of DDCC rules for medication recommendations involving disease pair (d_i, d_j). $M_{i\ j}$ represents the set of contraindicated medications corresponding to the disease pair (d_i, d_j), $\mathbb{I}(\cdot)$ is an indicator function, which takes the value of 1 when the recommended medications contains the contraindicated medication m_k, and 0 otherwise. $N_{i\ j}$ is the total number of patients diagnosed with both d_i and d_j.

6.3. Baselines

This study compares KATMed with three categories of representative baselines:

(i) Shallow non-temporal models, (ii) Deep sequential models without explicit external knowledge and (iii) Knowledge-enhanced models. This grouping helps clarify the distinct purpose of each baseline category. Below is a detailed description of each category:

(i) Shallow Non-Temporal Models.

These models do not explicitly account for the temporal structure of EHR data and are based on simple algorithms. They represent the baseline performance for static, non-sequential data processing.

• **LR (Logistic Regression)** is a shallow linear model widely used in multi-label classification. It takes the aggregated multi-hot vectors of multi-visit EHRs as input and directly predicts medication labels based on weighted features. This model serves as a strong non-temporal, non-contextual baseline.

(ii) Deep sequential models without explicit external knowledge.

These models incorporate the temporal structure of EHR data, learning from the sequential visits to make predictions. However, they do not integrate any external clinical knowledge (e.g., disease–procedure associations or contraindication rules).

• **LEAP**[2] is an instance-based model designed for single-visit data. It uses content-based attention to link disease codes to medications and employs a recurrent decoder to model label dependencies. This model captures code-to-medication mappings without temporal modeling and thus serves as a deep, data-driven baseline without explicit external knowledge.

• **RETAIN**[36] introduces a reverse-time attention mechanism to assign weights to prior visits and variables, mimicking physician reasoning. It focuses on visit-level temporal dynamics for interpretable recommendations and serves as a representative deep sequential baseline without explicit external knowledge.

• **MICRON**[21] models longitudinal prescriptions as soft constraints in a recurrent residual architecture. It incorporates past medications to enhance safe multi-visit recommendations and represents another deep sequential baseline without explicit external knowledge.

• **COGNet**[37] applies a copy-or-predict mechanism to combine current visit data with historical medication patterns. It supports personalized drug sequence generation guided by past records and serves as a deep sequential baseline emphasizing historical prescription patterns without explicit external knowledge.

(iii) Knowledge-Enhanced Models.

These models integrate clinical knowledge, such as disease–drug indications and drug–drug contraindications, into the recommendation process. They aim to enhance the predictive accuracy and safety of medication recommendations.

• **GAMENet**[22] uses graph-augmented memory networks to incorporate DDI knowledge into the recommendation process. It jointly encodes EHR sequences and medication co-occurrence graphs for safer predictions, representing a knowledge-enhanced DDI-aware baseline.

• **SafeDrug**[19] leverages dual graph neural networks over drug molecular structures and DDI graphs. It enhances medication safety by penalizing incompatible drug pairs using structural pharmacology and serves as a knowledge-enhanced baseline focused on DDI-aware safety.

• **G-Bert**[38] integrates medical ontology embeddings and graph-based token semantics into a BERT-like encoder. It performs self-

supervised pre-training and models multi-visit EHRs using contextualized graph-enhanced embeddings, representing a knowledge-enhanced pre-trained Transformer baseline.

- **MoleRec**[35] employs molecular graph learning to derive drug representations from substructures and recommend personalized combinations. It integrates chemical structure-level signals with medication selection and serves as a knowledge-enhanced baseline emphasizing molecular structure information.
- **StratMed**[39] uses a hierarchical attention network to model temporal diagnostic-treatment patterns. It improves medication safety through stratified modeling of long-term EHR sequences and can be viewed as a deep sequential baseline that partially integrates structured prior knowledge.
- **CausalMed**[40] is a causality-driven framework that captures direct causal pathways between clinical variables and prescriptions. It enhances interpretability and robustness by simulating patient trajectory evolution, representing a deep sequential baseline with an explicit causal modeling component but without DDCC-specific constraints.
- **MEGACare**[20] builds a multi-view hypergraph from heterogeneous medical knowledge sources, including code descriptions, ontology graphs, and molecular data. By leveraging hypergraph learning, it achieves high interpretability and improves both disease and medication prediction tasks, serving as a strong knowledge-enhanced baseline.

7. Experimental Setup

All experiments were implemented in PyTorch2.2.0 with CUDA12.1and executed on a workstation equipped with an NVIDIA RTX 3060 GPU (12 GB VRAM), 32 GB RAM, and Intel Core i7-12700 CPU. The experimental environment and all library dependencies were version-pinned to ensure reproducibility.

7.1. Data splits and reproducibility

For both MIMIC-III and MIMIC-IV datasets, this study followed the conventional 7:1:2 ratio for the training, validation, and test sets, stratified by patients rather than visits to prevent data leakage across subsets. All random operations, including sampling, shuffling, and weight initialization, were seeded with seed = 2024. Data preprocessing scripts (code normalization, tokenization, and visit ordering) are deterministic, ensuring that the same raw EHR data produce identical serialized input sequences across runs.

7.2. Pretraining and optimization

KATMed was pre-trained on the training set using a two-stage sequential strategy: first on the **ISP (Interaction Structure Prediction)** task and then on the **MSR (Medication Sequence Reconstruction)** task, with 40 epochs for each pretraining phase. The Transformer encoder comprises 4 layers, each with 4 self-attention heads, an embedding dimension of 512, and feed-forward dimension of 2048. The optimizer is AdamW with a learning rate of 1×10^{-5}, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and weight decay of 0.1. Early stopping is triggered if the validation F1 score does not improve for 5 consecutive epochs. Batch size is 32 for all models. The learning rate scheduler follows a linear warm-up over the first 5 epochs.

7.2.1. Loss weighting and hyperparameters

The initial balancing coefficient α in Eq. (15) was set to 0.75. Weights for loss components were initialized as $\varphi = 0.05$, $\lambda = 0.6$, and $\beta = 0.5$, determined by grid search on the validation set. To ensure fairness, all baseline models (e.g., LEAP, GAMENet, SafeDrug, MICRON) were trained with their publicly available implementations under identical data preprocessing, batch size, and evaluation metrics. Hyperparameter search followed the same validation-based tuning strategy across models.

7.2.2. Evaluation protocol

We report average results over five independent runs with different initialization seeds to reduce stochastic variance. Evaluation metrics include Jaccard similarity, F1-score, PRAUC, and DDCC-Violation Rate (DDCC-VR). Each metric is computed using scikit-learn1.5 and averaged across all test patients. For fairness, the same test partition is used across all compared methods. All experiments are logged and tracked via MLflow, enabling reproducibility of model checkpoints, configurations, and metrics.

8. Results

The results on single-visit and multi-visit settings are presented in Table 6 and Table 7, respectively. Below we analyze the models from three perspectives: accuracy (F1 score), safety (DDCC-VR), and prescribing behavior (#MED).

8.1. Single-visit-based baselines are limited by lack of temporal or semantic modelling

LR, as a linear model without structural encoding, consistently performs poorly across all metrics. Its F1 scores are only 0.6423 (MIMIC-III) and 0.5459 (MIMIC-IV), and its DDCC-VR reaches 0.0797—the highest among all models—highlighting a significant safety concern. **LEAP**, designed for single-visit settings, also shows subpar F1 performance (0.6154 and 0.5437), and fails to benefit much from multi-visit input, confirming its lack of temporal modeling capability. These results emphasize the critical importance of temporal modeling based on multi-visit records for medication recommendation. When relying solely on single-visit records, deep learning models may fail to demonstrate their advantages and could even underperform compared to traditional shallow models.

8.2. Multi-visit-based baselines benefit from longitudinal data but lack explicit knowledge safety

A comparison of Tables 6 and 7reveals that, **RETAIN**, **MICRON**, **COGNet** and **StratMed** demonstrate notable F1 gains in multi-visit settings compared to single-visit ones. For instance, **StratMed**'s F1 improves from 0.6262 to 0.6542 in MIMIC-III and from 0.6022 to 0.6212 in MIMIC-IV. This affirms that modeling visit-level sequences can enhance recommendation accuracy. However, their DDCC-VR values remain relatively high (>0.06 in most cases), e.g., **COGNet** reaches 0.0846 in MIMIC-IV, indicating persistent safety risks due to lack of knowledge supervision. Moreover, #MED analysis shows some models, such as **COGNet** in Table 6 (averaging 25.05 medications), tend to overprescribe, reflecting imbalanced decision boundaries.

8.3. Knowledge-enhanced models improve accuracy but lack DDCC-level safety supervision

Table 6 and Table 7 show that all knowledge-enhanced models, including **GAMENet**, **SafeDrug**, **G-Bert**, **MoleRec**, **CausalMed**, and **MEGACare**, consistently outperform shallow baselines (e.g., LR and LEAP) in terms of F1 score across both single-visit and multi-visit settings. For instance, **MoleRec** achieves 0.6798 and 0.6018 in MIMIC-III and MIMIC-IV single-visit cases (Table 6), outperforming LR by over 3%. **SafeDrug** and **GAMENet** also show notable F1 gains, with **SafeDrug** reaching 0.6098 (MIMIC-IV) and **GAMENet** 0.6762 (MIMIC-III). **G-Bert**, **MEGACare**, and **CausalMed** similarly deliver higher F1 scores than non-temporal baselines, benefiting from contextualized or structured representations.

However, despite these improvements in predictive accuracy, these models do not explicitly incorporate DDCC rules into their training objectives. As a result, they struggle to suppress multi-way contraindications arising from comorbidities. This is reflected in their relatively high

Table 6
Performance comparison on single-visit records. RETAIN and G-Bert are excluded because they cannot generate recommendations for single-visit patients.

	MIMIC-III					MIMIC-IV				
Method	Jaccard	PRAUC	F1	DDCC-VR	#MED	Jaccard	PRAUC	F1	DDCC-VR	#MED
LR	0.4925	0.7511	0.6423	0.0797	16.85	0.4141	0.6594	0.5459	0.0729	9.84
LEAP	0.4526	0.6583	0.6154	0.0722	19.04	0.3907	0.5540	0.5437	0.0548	12.66
MICRON	0.5119	0.5190	0.6676	0.0677	20.94	0.4495	0.4353	0.6033	**0.0502**	14.50
COGNet	0.5109	0.6465	0.6676	0.0737	25.05	0.4373	0.6112	0.5850	0.0866	15.46
GAMENet	0.5210	0.7780	0.6762	0.0781	19.78	0.4401	0.6833	0.5933	0.0718	12.89
SafeDrug	0.5187	0.7646	0.6725	0.0701	21.32	0.4560	0.6858	0.6098	0.0689	14.07
MoleRec	0.5287	0.7812	0.6798	0.0678	22.12	0.4511	0.6458	0.6018	0.0694	14.14
StratMed	0.5012	0.7431	0.6262	0.0722	20.23	0.4923	0.6481	0.6022	0.0685	15.65
CausalMed	0.4988	0.5052	0.6168	0.0710	22.35	0.3645	0.5152	0.5054	0.0744	20.32
MEGACare	0.5012	0.6161	0.6403	0.0604	19.64	0.4845	0.6624	0.6123	0.0690	17.64
KATMed	**0.5564**	**0.8337**	**0.6841**	**0.0401**	19.75	**0.5086**	**0.7154**	**0.6184**	0.0511	18.54

Table 7
Performance comparison on multi-visit records.

	MIMIC-III					MIMIC-IV				
Method	Jaccard	PRAUC	F1	DDCC-VR	#MED	Jaccard	PRAUC	F1	DDCC-VR	#MED
LR	0.4925	0.7511	0.6423	0.0797	16.85	0.4141	0.6594	0.5459	0.0729	9.84
LEAP	0.4526	0.6583	0.6154	0.0722	19.04	0.3907	0.5540	0.5437	0.0548	12.66
RETAIN	0.4941	0.7465	0.6587	0.0785	19.36	0.4104	0.6168	0.5958	0.0812	17.46
MICRON	0.5188	0.6398	0.6753	0.0658	19.51	0.4555	0.5350	0.6081	0.0528	13.24
COGNet	0.5177	0.6360	0.6715	0.0757	24.95	0.4518	0.6359	0.6026	0.0846	15.70
GAMENet	0.5205	0.7768	0.6754	0.0774	19.52	0.4462	0.6923	0.5979	0.0807	12.60
SafeDrug	0.5246	0.7521	0.6468	0.0697	20.45	0.4478	0.6756	0.5928	0.0692	14.37
G-Bert	0.5037	0.7631	0.6617	0.0832	24.46	0.4292	0.6746	0.5822	0.0794	16.20
MoleRec	0.5214	0.7765	0.6631	0.0681	22.24	0.4594	0.6578	0.6118	0.0686	14.84
StratMed	0.5412	0.7521	0.6542	0.0622	20.13	0.4943	0.6672	0.6212	0.0613	14.95
CausalMed	0.5011	0.5172	0.6223	0.0694	22.25	0.3841	0.5274	0.5194	0.0731	20.12
MEGACare	0.5012	0.6161	0.6403	0.0604	19.64	0.4845	0.6624	0.6123	0.0690	17.64
KATMed	**0.5564**	**0.8337**	**0.6841**	**0.0401**	19.75	**0.5086**	**0.7154**	**0.6184**	**0.0511**	18.54

DDCC-VR values across datasets: **SafeDrug** and **GAMENet** reach 0.0689 and 0.0807, respectively, in MIMIC-IV (Tables 6 and 7); **G-Bert** and **CausalMed** exceed 0.07 in most settings; even the more balanced **MEGACare** attains only moderate safety levels (e.g., 0.0690 in MIMIC-IV multi-visit). These results highlight a key limitation of existing knowledge-enhanced models: while they leverage external graphs or ontologies to improve general prediction, they lack mechanisms to encode and enforce comorbidity-aware safety constraints such as DDCC rules.

8.4. Katmed achieves state-of-the-art performance across all metrics

Our proposed **KATMed** consistently outperforms all baselines in both single-visit and multi-visit settings. In terms of F1 score, KATMed achieves 0.6841 (MIMIC-III) and 0.6184 (MIMIC-IV), outperforming the strongest baselines (**MoleRec**, **StratMed**, **MEGACare**) by margins of 0.5–2.5 percentage points. Crucially, KATMed achieves the lowest DDCC-VR scores across both datasets: 0.0401 (MIMIC-III) and 0.0511 (MIMIC-IV), reflecting superior safety in handling DDCC. These results validate the design of KATMed, which uniquely integrates bidirectional clinical knowledge—positive disease-procedure associations and negative DDCC rules—enabling contraindication-aware medication recommendation in comorbidities.

8.5. Prescribing balance analysis via #MED

The #MED metric reflects how aggressively a model tends to recommend medications. **LR** shows conservative recommendations (9.84 in MIMIC-IV), while **COGNet** shows over-prescription (up to 25.05). In contrast, **KATMed** maintains a clinically reasonable balance (19.75 in MIMIC-III and 18.54 in MIMIC-IV), aligning closely with empirical statistics of real-world prescriptions.

8.6. Single-visit vs multi-visit setting comparison

Comparing across the two tables, most deep models (e.g., **MICRON**, **StratMed**) show F1 improvement in the multi-visit settings, confirming the necessity of leveraging historical visit information. However, KATMed already performs well in the single-visit setting and shows consistent improvement in the multi-visit case without requiring complex temporal modeling. This suggests that KATMed effectively captures patient context through its unified encoder and knowledge-enhanced representation.

These results support the following conclusions: (a) multi-visit modeling is essential for long-term accuracy; (b) deep learning models are significantly better than shallow ones, especially with sequential data; (c) knowledge-enhanced approaches, particularly those incorporating DDI and contraindication knowledge, are more robust and safer than data-only models; (d) our KATMed stands out as the only model that jointly considers comorbidity-aware safety and bidirectional clinical associations, achieving the best trade-off between accuracy, safety, and prescription balance.

8.7. Ablation study

The integration of diverse embedding strategies, pre-training tasks, and knowledge enhancement techniques constitutes the key innovation of KATMed. To evaluate their effectiveness, this study designs the following model variants for an ablation study:

•*KATMed-PT*(without ISP and MSR pretraining): This variant removes the two pre-training tasks from KATMed, resulting in random initialization of the embedding layer. It is used to evaluate the effectiveness of the self-supervised pre-training module.

•*KATMed-AE*(without disease–procedure knowledge E_{ass}): This variant removed association embedding from KETMed, so that the

patient's information representation can only rely on the token representations after separating the diseases and procedures. It is used to evaluate the effectiveness of knowledge enhancement regarding positive diseases-procedures associations.

•*KATMed-SE*: This variant removed the segment embedding, so that the patient's information representation can only rely on the token representation of association embedding. It is used to evaluate the effectiveness of segment embedding.

•*KATMed-CL*(without DDCC constraints L_{constr}): This variant removed the contraindication constraints. It is used to evaluate the effectiveness of knowledge enhancement regarding DDCC.

•$KATMed_{AE\&-CL}$ (plain Transformer baseline without knowledge augmentation). This variant simultaneously disables the disease–procedure association embedding E_{ass} and the DDCC constraint loss L_{constr}, while keeping the Transformer backbone architecture, input structuring and training protocol unchanged. It therefore serves as a plain Transformer-based baseline without any external clinical knowledge augmentation.

Ablation experiments were conducted on the MIMIC-III dataset. Table 8 presents the results of five model variants:

• **KATMed-PT** leads to a noticeable decline in F1 (↓1.07%) and PRAUC (↓4.81%), indicating that pre-training sig- nificantly enhances the model's ability to capture generalizable clinical patterns and improves overall representation quality. Although DDCC-VR remains low (0.0404), the model's predictive accuracy suffers without pretraining-based initialization.

• **KATMed-AE** exhibits the most severe drop in F1 score (↓3.29%) and Jaccard (↓3.4%), reflecting a sharp decline in structural semantic modeling capability. Without positive diseases-procedures associations, the model struggles to align diagnostic and procedural signals, resulting in significantly weaker patient representations.

• **KATMed-SE** has a moderate F1 drop (↓1.47%), demonstrating that segment embedding improves input structure learning of model. The accuracy metrics show that even subtle architectural features like segment identifiers contribute positively to the final performance.

• **KATMed-CL** retains relatively high predictive accuracy (F1 only ↓0.30%), which is expected given the low prevalence of contraindicated drug-disease pairs. However, it results in a severe degradation of DDCC-VR (from 0.0401 to 0.0684), confirming that although such constraints may not affect general prediction accuracy, they are critical for ensuring clinical safety.

• **KATMed-AE&-CL** shows a clear loss of performance (F1 ↓3.07%) together with a marked deterioration in safety (DDCC-VR ↑ from 0.0401 to 0.0702). This indicates that, once external clinical knowledge is removed from both the representation and the optimization objectives, the model can still fit temporal patterns in the data, but its ability to make safe and clinically aligned recommendations is substantially weakened, underscoring the importance of knowledge-enhanced design in KATMed.

Among all modules, disease–procedure association embedding is most critical for improving predictive accuracy, while the DDCC contraindication constraints play a pivotal role in controlling safety-related violations. The pre-training tasks enhance representational robustness, and segment embeddings help encode structural input semantics. These findings affirm the necessity of KATMed's integrated design for safe and accurate medication recommendation.

Table 8
Ablation study on MIMIC-III dataset.

Model	MIMIC-III				
	Jaccard	PRAUC	F1	DDCC-VR	#MED
KATMed-PT	0.5382	0.7856	0.6768	0.0404	18.85
KATMed-AE	0.5245	0.8010	0.6512	0.0425	19.75
KATMed-SE	0.5507	0.8212	0.6694	0.0404	18.99
KATMed-CL	0.5502	0.8294	0.6811	0.0684	19.48
$KATMed_{AE\&-CL}$	0.5278	0.8073	0.6534	0.0702	19.62
KATMed	**0.5564**	**0.8337**	**0.6841**	**0.0401**	19.75

To further understand the role of pre-training, Fig. 4 compares model performance under varying epochs and dimensions.

We observe that the ISP task benefits from early-stage training (5–10 epochs) and shows diminishing returns thereafter. In contrast, the MSR task benefits from deeper contextual encoding with longer training and higher embedding dimensions (up to 512), especially improving F1 and PRAUC. These patterns explain the performance gap between KATMed and *KATMed-PT*, demonstrating that both tasks provide complementary gains and contributes to robust model initialization.

In addition, hyperparameter sensitivity is visualized in Fig. 5. Among the four parameters, over-weighting the constraint loss ($\lambda > 0.6$) suppresses PRAUC, confirming that excessive emphasis on DDCC rules can hinder generalization. Similarly, optimal values are observed near $\varphi = 0.05$, $\beta = 0.5$, and $\mu = 0.5$, ensuring that positive constraints, loss balancing, and output thresholding collectively maintain high accuracy and safety. These experiments confirm the necessity of a balanced integration of pre-training, knowledge embedding, and contraindication constraints to optimize both clinical performance and medication safety.

Cross-dataset and external validation

To further evaluate the robustness and generalizability of KATMed, this study conduct two additional experiments: a cross-version, cross-dataset evaluation between MIMIC-III and MIMIC-IV, and a partial external validation on the multi-center eICU cohort.

Cross-dataset evaluation: training on MIMIC-III → testing on MIMIC-IV. Because MIMIC-IV is an updated and expanded release that includes many of the patients and admissions from MIMIC-III, a naive train-on-III/test-on-IV setting may introduce patient-level overlap. To avoid this, this study first identify overlapping admissions using subject and admission identifiers, then exclude these overlapping admissions from the MIMIC-IV test set, and finally construct a non-overlapping MIMIC-IV test subset that remains representative of the overall case mix. All hyperparameters, preprocessing steps, and evaluation metrics are kept identical to the in-domain experiments.

Quantitatively, as summarized in Table 9, all models exhibit a moderate degradation when moving from in-domain evaluation on MIMIC-IV (Table 7) to the cross-dataset setting. For example, KATMed's F1 decreases from 0.6184 to 0.6031 (↓1.53%) and its PRAUC from 0.7154 to 0.6992, whereas StratMed drops from 0.6212 to 0.6020 in F1 and from 0.6672 to 0.6451 in PRAUC. Despite this performance drop, KATMed still achieves the best overall accuracy (Jaccard = 0.4950, F1 = 0.6031, PRAUC = 0.6992) and maintains a lower DDCC-VR (0.0534) than knowledge-agnostic baselines such as GAMENet and SafeDrug. Moreover, the relative performance drop of KATMed is smaller than that of several baselines, suggesting that the incorporation of disease–procedure associations and DDCC rules improves robustness under distribution shift between MIMIC-III and MIMIC-IV.

Partial external validation on a multi-center cohort (eICU). To examine external generalization beyond a single academic medical center, this study treat the constructed eICU cohort as an external test set and reuse the KATMed model trained on the combined MIMIC-III and MIMIC-IV training cohorts, without any fine-tuning on eICU. The same set of baseline models is evaluated under this "train-on-MIMIC, test-on-eICU" setting using identical preprocessing and evaluation metrics (F1, Jaccard, PRAUC, DDCC-VR, #MED) as in the main experiments.

As shown in Table 10, all models show a noticeable degradation on eICU compared with their in-domain performance on MIMIC-IV, owing to differences in coding distributions, practice patterns, and data completeness across institutions. For instance, KATMed's F1 decreases from 0.6184 (MIMIC-IV, Table 7) to 0.5912 on eICU, and its Jaccard from 0.5086 to 0.4385. Nonetheless, KATMed still achieves the best external performance among all compared methods (Jaccard = 0.4385, F1 = 0.5912, PRAUC = 0.6683) and maintains a lower DDCC-VR (0.0621) than knowledge-agnostic baselines such as GAMENet and RETAIN. These results suggest that the bidirectional knowledge

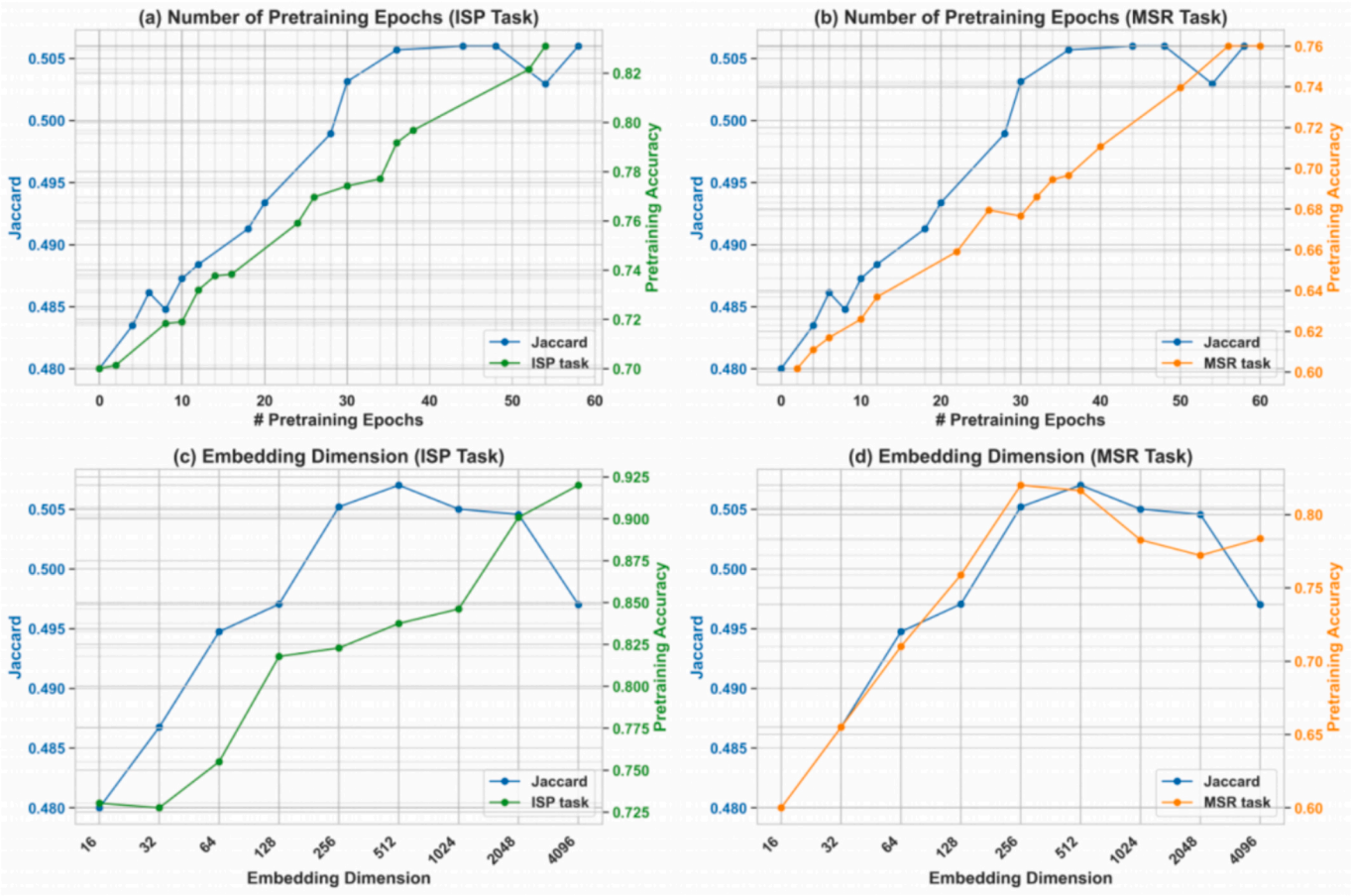


Fig. 4. Effect of Pretraining Tasks and Embedding Dimensions on Representation Quality.

components in KATMed (positive disease–procedure associations and negative DDCC rules) confer safety and robustness benefits that generalize, at least partially, to a multi-center ICU cohort with different data characteristics.

The detailed cross-dataset results are reported in Table 9, and the external validation results on eICU are summarized in Table 10.

9. Limitations and future work

Limitations and future work. This study has several limitations. First, although KATMed was evaluated on two publicly available critical care datasets (MIMIC-III and MIMIC-IV) and partially externally validated on the multi-center eICU Collaborative Research Database, all cohorts still represent ICU populations from hospitals in the United States; practice patterns, case mix and coding conventions may differ substantially in other hospitals, healthcare systems or countries. In addition, our current definition of a visit at the admission (HADM_ID) level, which merges ICU stays within a single admission, may smooth over intra-admission temporal nuances such as early versus late ICU phases. Even with the added cross-version (MIMIC-III → MIMIC-IV) and external (eICU) experiments, broader multi-center and multi-country validation and prospective clinical studies will therefore be needed to fully establish the generalizability and clinical impact of the proposed framework. Second, the disease–procedure associations and DDCC rules are mainly derived from clinical guidelines, textbooks and reference materials, which, despite being authoritative, may be incomplete or lag behind emerging evidence and newly approved therapies. Incompleteness or delayed updates in these sources could lead to missing or outdated constraints. Third, the current implementation relies on structured EHR codes (diseases, procedures and medications) and does not yet incorporate unstructured clinical text, medical images or genomic data, which may provide additional context for both prediction and interpretability. Future work will explore integrating multimodal clinical data and developing mechanisms to dynamically update the constraint knowledge base as new data and knowledge become available, further enhancing the robustness, safety and applicability of KATMed in real-world settings.

10. Conclusion

The proposed KATMed model in this study overcomes the challenges in medication recommendation modeling caused by the long-tail distribution characteristics of DDCC through the integration of bidirectional clinical associations. Positive association embedding leverages dynamically weighted matrices derived from clinical guidelines to enhance disease-procedure alignment, while negative association constraints explicitly suppress contraindicated drugs through logical loss functions, surpassing the limitations of existing DDI-centric knowledge integration approaches. Experiments demonstrate that bidirectional knowledge enhancement significantly improves both accuracy and safety across single-visit and multi-visit scenarios. Future research will focus on integrating diverse clinical data types (e.g., imaging and genomics) and developing automated constraint generation mechanisms to further enhance interpretability and adaptability in complex clinical decision-making.

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Institutional review board statement

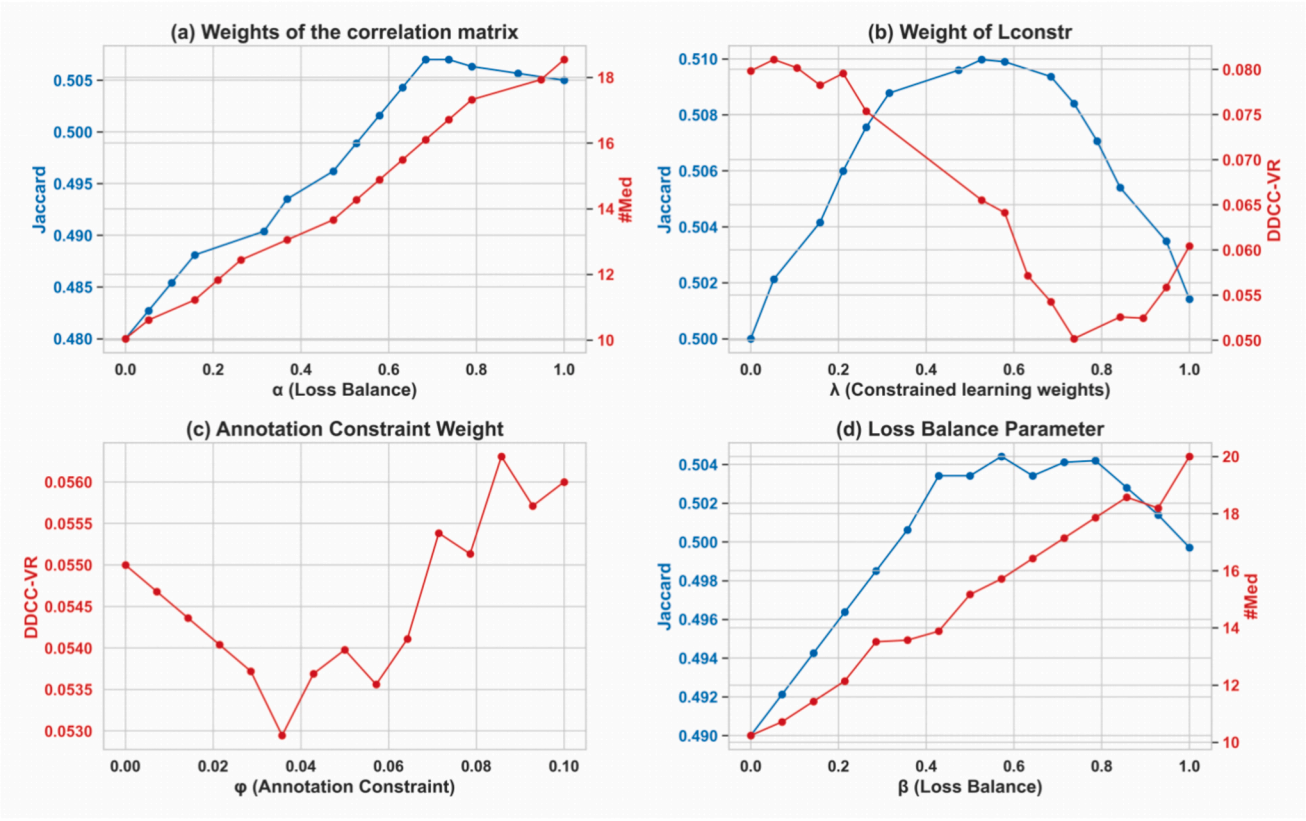


Fig. 5. Effect of Hyperparameter Variations on Model Performance and Safety.

Table 9
Cross-dataset evaluation (train on MIMIC-III, test on non-overlapping MIMIC-IV, multi-visit setting).

Method	Jaccard	PRAUC	F1	DDCC-VR	#MED
RETAIN	0.3982	0.5954	0.5793	0.0841	17.20
MICRON	0.4421	0.5182	0.5930	0.0553	13.05
GAMENet	0.4324	0.6671	0.5824	0.0830	12.52
SafeDrug	0.4352	0.6523	0.5791	0.0715	14.10
MEGACare	0.4703	0.6402	0.5982	0.0712	17.05
StratMed	0.4801	0.6451	0.6020	0.0634	14.82
KATMed	**0.4950**	**0.6992**	**0.6031**	**0.0534**	**18.30**

Table 10
External validation on the eICU cohort (train on MIMIC-III + IV, test on eICU, multi-visit setting).

Method	Jaccard	PRAUC	F1	DDCC-VR	#MED
RETAIN	0.3824	0.5895	0.5604	0.0903	16.80
MICRON	0.4012	0.5021	0.5682	0.0704	12.90
GAMENet	0.3950	0.6352	0.5623	0.0931	12.34
SafeDrug	0.3985	0.6204	0.5592	0.0825	13.82
MEGACare	0.4153	0.6405	0.5751	0.0783	16.92
StratMed	0.4221	0.6450	0.5784	0.0745	14.60
KATMed	**0.4385**	**0.6683**	**0.5912**	**0.0621**	**17.80**

The Massachusetts Institute of Technology (Cambridge, MA, USA) and the Institutional Review Boards of Beth Israel Deaconess Medical Center (Boston, MA, USA) have approved the data collection and the use of MIMIC-III and MIMIC- IV for research purposes and granted waiver of informed consent. The author Ziqian Qiao obtained the certificate (no: 61386984) and the permission to use the MIMIC-III and MIMIC-IV database after completing the web-based course. https://physionet.org/content/mimiciii/1.4/ and https://physionet.org/content/mimiciv/3.1/ (accessed on 21 February 2024).

CRediT authorship contribution statement

Ziqian Qiao: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Shaofu Lin:** Writing – review & editing, Supervision, Funding acquisition. **Jiatong Fan:** Investigation, Formal analysis. **Jianhui Chen:** Conceptualization, Methodology, Writing – review & editing, Project administration. **Zhiyi Tang:** Visualization, Validation, Methodology. **Zitong Zhang:** Validation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbi.2026.104991.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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